

CSCI 2503 – Survey of Programming Language
Georgia Southern University
Department of Computer Science
Summer 2026

Assignment 5

Point Value: 20 points

Due: Thursday June 25, 2026, 11:59pm

Description

1. **C++ Structs** – Complete the following requirements.

Recreate the provided Java class from the last assignment (Fan.java) and main method in C++ using a struct to implement the Fan. Make sure the default values are set upon initialization (this is discussed in Chapter 13.9 at LearnCPP.com). Consider if every piece of the class-based implementation can be implemented as a struct.

Consider if data field encapsulation still works with a struct.

This can be done in a single file if you want. Consider where any function implementations need to be placed in regard to the main method.

QUESTION(S): Compare and contrast this to writing a class for a Fan like you did in C#. In other words, what things are easier/harder to do? Do you lose access to possibly wanted functionality when using a non-class based approach? Is there anything you can do with a struct's field that could be prevented with a class's data member? Is there any part of the class that you could not replicate?

Create a Word document and submit the answers to these questions. Create a zip file and include your working C++ struct and test program in this zip file (use this zip file for any code requested in this document).

Take a screenshot of your output and include this in your zip file.

2. **Learning Methodology** – Complete the following requirements.

Consider working with a programming language that closely mirrors one you already know.

QUESTION(S): In your opinion, did you find it easier or quicker to start learning C and C++ (and even C#), considering your experience in Java, when compared to picking up a completely new language like Python or later Rust? Justify why you personally feel the way you do about the approach.

Add the answers to the above questions in the Word document created in Problem 1.

3. Convert your Word document to a PDF before submitting it. This PDF should **not** be added to the code/screenshot zip file.
4. Submit your **PDF** and **zip file** to the correct Folio Assignment dropbox. **Documents that are not in the requested format will not be graded and receive a grade of 0.**

Deliverables

Assignment 5 is to be individual work. Submit the required materials by the specified due date to the correct Folio dropbox.

Last modified: June 14, 2026